SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**

Question Count: 4

Code of Conduct

I Santhosh kumar.M(192511017)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Library Book Reservation System – SaaS

Aim

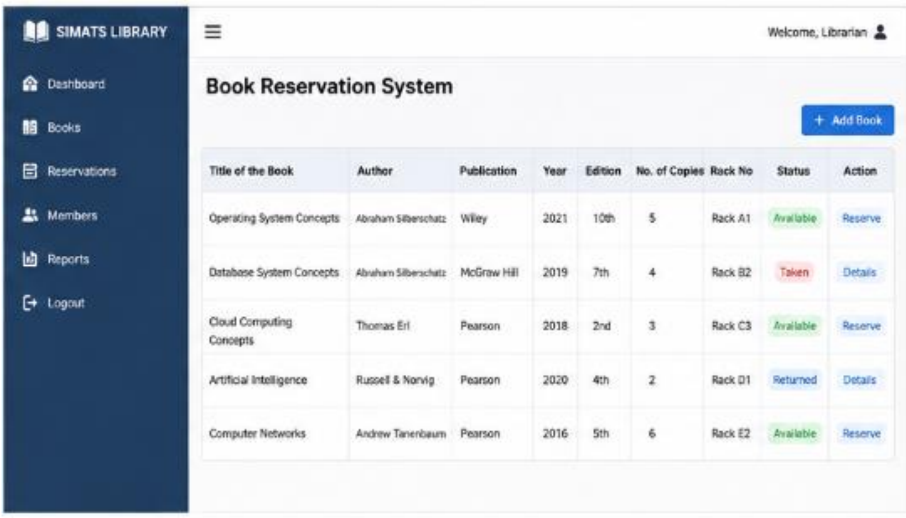
To create a simple cloud-based **Library Book Reservation System** using a Cloud Service Provider and demonstrate **Software as a Service (SaaS)**.

Procedure

1. Create an account with a cloud service provider.
2. Create a cloud database for library book details.
3. Add the required fields such as title, author, publication, year, edition, copies and rack number.
4. Create a simple application interface for entering and viewing book details.
5. Add the **Taken/Returned** status for each book.
6. Provide a reservation option for users.
7. Deploy the application as a cloud-based service.
8. Test the application by adding, searching and reserving books.

Result

Thus, a **cloud-based Library Book Reservation System** was created and successfully demonstrated as a **SaaS application**.



Experiment 2: Payroll Processing System – SaaS

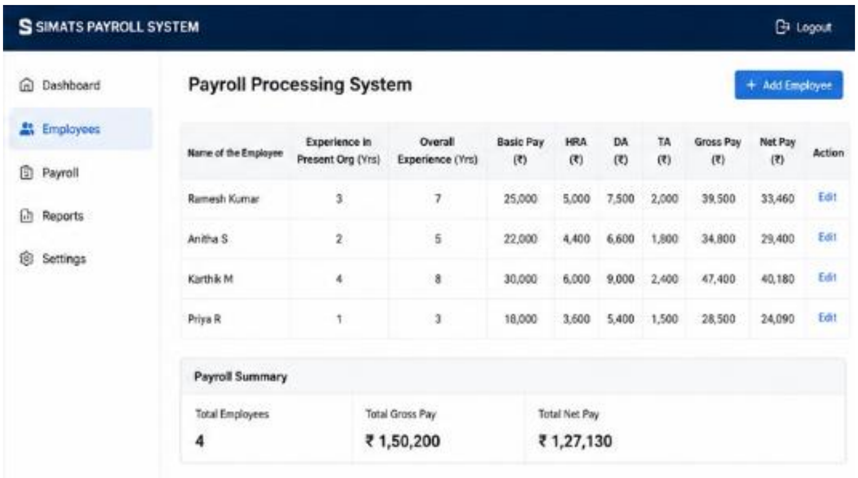
Aim

To create a simple cloud-based **Payroll Processing System** using a Cloud Service
Procedure

1. Create a cloud-based database for employee details.
2. Add employee information and salary-related fields.
3. Create an application interface for entering employee details.
4. Configure formulas for calculating gross pay and net pay.
5. Store the payroll information in the cloud.
6. Display the calculated salary details to the user.
7. Test the application using sample employee data.
8. Deploy and access the application through the cloud.

Result

Thus, a **cloud-based Payroll Processing System** was created and successfully demonstrated as a **SaaS application**.



Experiment 3: Creating a Virtual Machine Using Type-2 Hypervisor

Aim

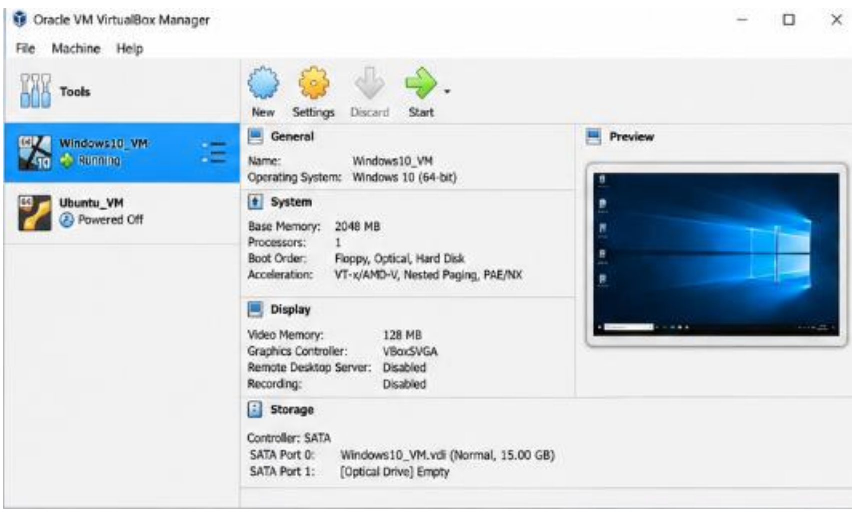
To demonstrate virtualization by installing a **Type-2 Hypervisor** and creating a virtual machine with **1 CPU, 2 GB RAM and 15 GB storage**.

Procedure

1. Install a Type-2 hypervisor such as VirtualBox on the host computer.
2. Open the virtualization software.
3. Select **Create/New Virtual Machine**.
4. Enter a suitable name for the VM.
5. Select the downloaded Windows/Linux ISO image.
6. Allocate **1 CPU core**.
7. Allocate **2 GB RAM**.
8. Create a virtual hard disk with **15 GB storage**.
9. Complete the VM configuration.
10. Start the VM and install the guest operating system.
11. Verify that the guest OS runs successfully inside the virtual machine.

Result

Thus, a virtual machine was successfully created using a **Type-2 Hypervisor** with **1 CPU, 2 GB RAM and 15 GB storage**.



Experiment 4: Cloning a Virtual Machine

Aim

To demonstrate virtualization by creating a **clone of an existing virtual machine** and testing the cloned VM.

Procedure

1. Open the installed Type-2 Hypervisor.
2. Select the previously created virtual machine.
3. Make sure the original VM is powered off.
4. Select the **Clone** option.
5. Enter a name for the cloned VM.
6. Select **Full Clone** to create an independent copy.
7. Start the cloned virtual machine.
8. Verify that the operating system and previously installed applications are available.
9. Test the cloned VM by opening files or applications from the previous version.
10. Confirm that the cloned VM works independently from the original VM.

Result

Thus, the existing virtual machine was successfully **cloned**, and the cloned VM was tested and found to work correctly with the previous configuration.